Unconstrained Optimization

ECON 441: Introduction to Mathematical Economics
Lecture 14

Div Bhagia

Optimization

How many hours to work each week?

Utility from consumption (C) and leisure (L):

$$U(C, L)$$

Leisure is the difference between total hours (T) and hours worked (H):

$$L = T - H$$

Constraint:

$$wH = pC$$

where w and p denote the wage and the price, respectively.

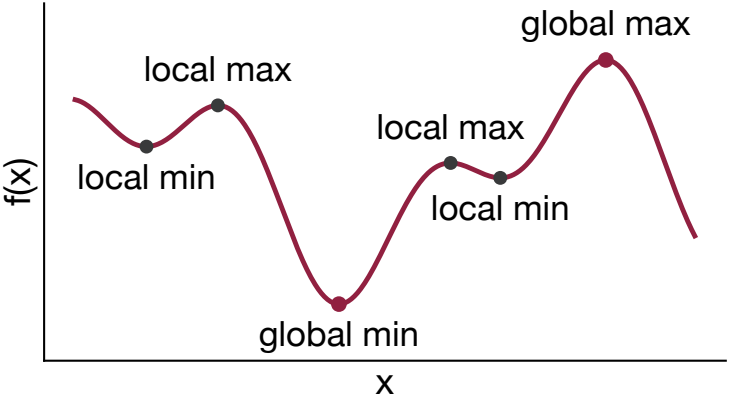
Optimization in Economics

- Consumers choose consumption to **maximize** utility
- Firms choose output to **maximize** profit, or inputs to **minimize** cost
- Governments choose policy to maximize welfare, and so on

Today: the choice variable is a single number x and there are no constraints.

$$\max_x f(x) \qquad \text{or} \qquad \min_x f(x)$$

Global vs Local Extrema



Critical Points

We limit ourselves to functions that are **continuously differentiable**: f is continuous and has a continuous derivative.

$$y = f(x)$$

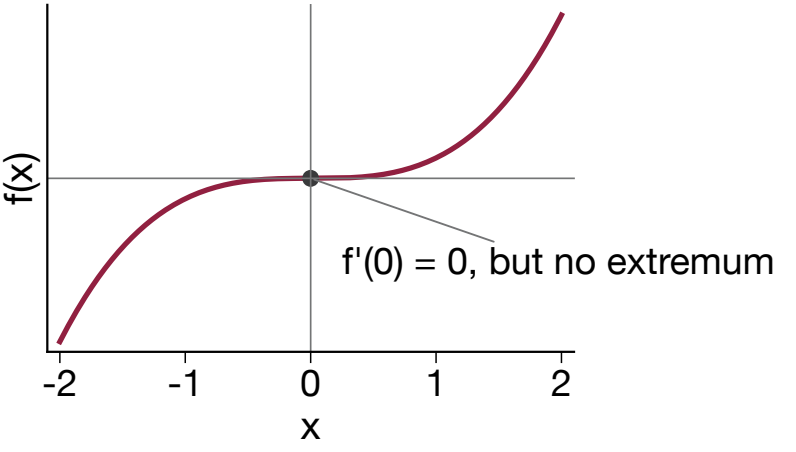
A **necessary** condition for x_0 to be an interior extremum:

$$f'(x_0) = 0$$

x_0 is called a **critical** (or **stationary**) point.

Why is this not sufficient?

Inflection Point

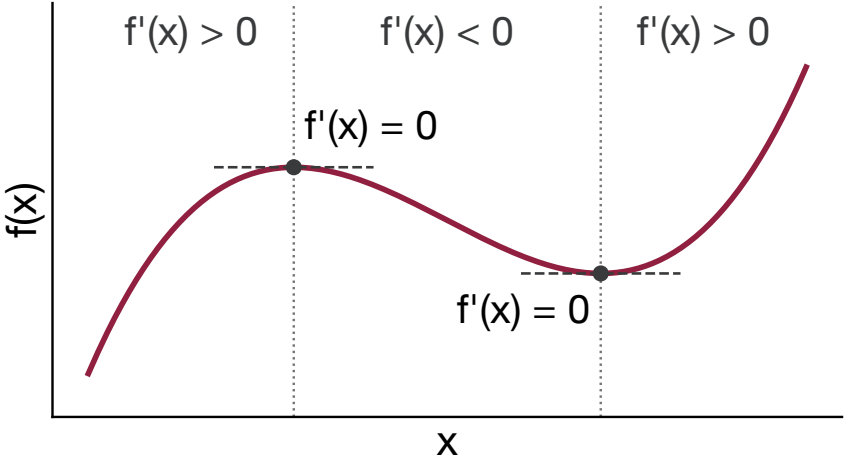


First-Derivative Test

Suppose that $f'(x_0) = 0$. Then $f(x_0)$ is a

1. **maximum** if $f'(x)$ goes from $+$ to $-$ in the immediate neighborhood of x_0
2. **minimum** if $f'(x)$ goes from $-$ to $+$ in the immediate neighborhood of x_0
3. **not an extremum** if $f'(x)$ has the same sign on both sides of x_0

First-Derivative Test



Example

Find the critical points of

$$f(x) = x^2 - 24x + 36$$

and determine whether each one is a maximum or a minimum.

Example

Find the critical points of

$$f(x) = 2x^3 - 9x^2 + 12x$$

and determine whether each one is a maximum or a minimum.

Example: Average and Marginal Cost

Total cost: $TC = C(Q)$

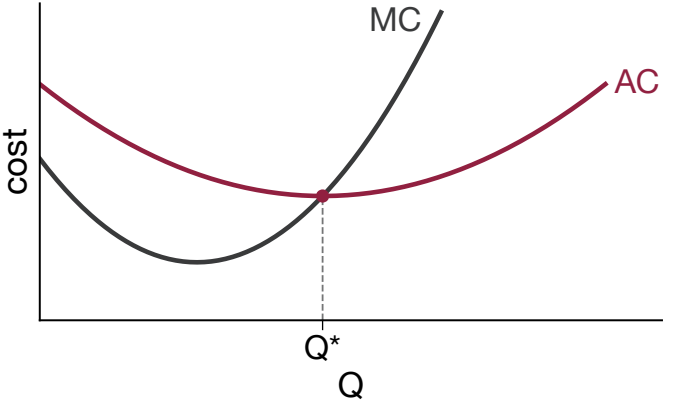
Marginal cost: $MC = C'(Q)$

Average cost:

$$AC = \frac{C(Q)}{Q}$$

At what quantity is AC the lowest?

Average and Marginal Cost



Example: Profit Maximization

A firm's profit is revenue minus cost:

$$\pi(Q) = R(Q) - C(Q)$$

What condition must hold at the profit-maximizing output Q^*?

Does every Q that satisfies this condition maximize profit?

Homework Questions

- Exercise 9.2: 1 (c), 2 (a), 3, 4

Textbook reference: 9.1, 9.2